functional

$g : \mathcal{F} \to \mathbb{R}$

$$\begin{cases} \theta(\underset{\smile}{f}) = f(0) \\ \theta(f) = \int_{-\infty}^{+\infty} x^n f(x)\, dx \end{cases}$$

Goal: How to find a lower bound for minimax error rate of a functional.

Def: Lipschitz Constant of a family of density w.r.t Hellinger dist:

$$\omega(\epsilon; \theta, \mathcal{F}) := \sup_{\substack{f, g \in \mathcal{F} \\ \to H^2(f \| g) \leq \epsilon^2}} |\theta(f) - \theta(g)|$$

Subgoal: Find m in terms of ω.

$$\underset{\sim}{m} \geq \frac{1}{2} \Phi(\delta) \left[1 - \| P^n_{\theta_1} - P^n_{\theta_2} \|_{TV} \right]$$

$\{\theta_1, \theta_2\}$ are 2δ-sep.

$\Phi(t)$ e.g. t^2

$$\rightarrow \underset{\sim}{\| P^n_{\theta_1} - P^n_{\theta_2} \|^2_{TV}} \leq H^2(P^n_{\theta_1} \| P^n_{\theta_2}) \leq n \underbrace{H^2(P_{\theta_1} \| P_{\theta_2})} \leq n \cdot \frac{1}{4n} = \underset{\sim}{\frac{1}{4}}$$

$f, g \in \mathcal{F}$ that achieves the sup in def. of $\omega(\cdot)$.

$$\epsilon^2 = \frac{1}{4n} \qquad \omega(\epsilon) := \sup_{\substack{f, g \in \mathcal{F} \\ \underbrace{H^2(f \| g) \leq \epsilon^2}}} |\theta(f) - \theta(g)|$$

with this selection of f & g :

$$2\delta = |\theta(f) - \theta(g)| = \omega(\epsilon) \longrightarrow \delta = \frac{\omega(\epsilon)}{2}$$

$$m \geq \frac{1}{2}\Phi(\delta)\cdot\Big[1 - \underbrace{\|P_f^n - P_g^n\|_{TV}}_{\leq \frac{1}{2}}\Big] \geq \frac{1}{4}\Phi(\delta) = \frac{1}{4}\Phi\Big(\frac{\omega(\epsilon)}{2}\Big)$$

$$= \underbrace{\frac{1}{4}\Phi\Big(\frac{1}{2}\omega\Big(\frac{1}{2\sqrt{n}}, \theta, \mathcal{F}\Big)\Big)}$$

E.g. Consider all densities on $[-1/2, 1/2]$, that are 1-Lipschitz

$\forall x, \Delta \quad |f(x+\Delta) - f(x)| \leq |\Delta|$. Suppose that $\underline{f(0)}$ is to be

estimated by a set n iid samples from $f(\cdot)$.

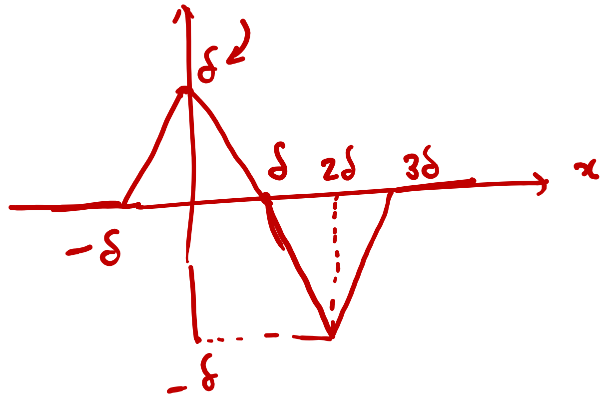
Find m.

Sol. $f_0(x) = 1 \quad x \in [-1/2, 1/2]$

$g(x) = f_0(x) + \varphi(x)$

$$\int_{-1/2}^{1/2} \varphi(x)\,dx = 0$$

$\varphi(x) \qquad \delta \in [0, 1/6]$



Note that f_0 & g are both 1-Lipschitz.

$$|\theta(f_0) - \theta(g)| = |\underbrace{f_0(0)}_{1} - \underbrace{g(0)}_{1+\delta}| = \delta$$

$$H^2(f_0 \| g) = \int (\sqrt{f_0} - \sqrt{g})^2 dx = 2 - 2\int \sqrt{f_0 g}\, dx$$

$$\tfrac{1}{2} H^2(f_0 \| g) = 1 - \int \sqrt{f_0 g}\, dx = 1 - \int_{-1/2}^{1/2} \sqrt{1 + \varphi(x)}\, dx$$

$$\psi(u) = \sqrt{1+u}$$

$$\frac{1}{2} H^2(f_0 \| g) = 1 - \int_{-1/2}^{1/2} \sqrt{1+\phi(t)}\, dt$$

$$= \int_{-1/2}^{1/2} \{ \underbrace{\psi(0)}_{\sim} - \psi(\phi(t)) \}\, dt$$

$0 \leq c \leq u$

$\exists c$

$$\psi(u) = \psi(0) + \psi'(0) u + \frac{\psi''(c)}{2} u^2$$

Intermediate Function theorem

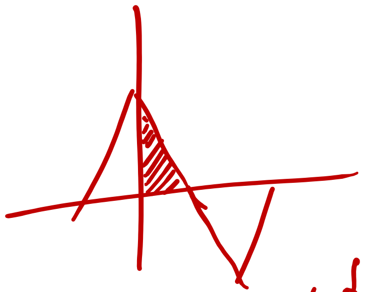
$$\frac{1}{2} H^2(f_0 \| g) = \int_{-1/2}^{1/2} \{ \cancel{\psi(0)} - \cancel{\psi(0)} - \underbrace{\psi'(0)\, \phi(t)}_{\text{Zero integral}} \stackrel{?}{=} \frac{\psi''(c)}{2} \phi^2(t) \} dt$$

$$\psi'(u) = \frac{1}{2\sqrt{1+u}} \qquad \psi''(u) = \frac{-1}{4(1+u)^{3/2}}$$

$|\psi''(c)| \leq 1/4$

$0 \leq c \leq 1/2$

$$\Rightarrow \quad \tfrac{1}{2} H^2(f_0 \| g) \le \int_{-1/2}^{1/2} \frac{1}{2\cdot 4} \phi^2(t)\, dt$$



$$\frac{1}{2}\int_{-1/2}^{1/2} \phi^2(t)\, dt = \frac{4}{2}\int_0^{\delta} (\delta - t)^2\, dt = -\frac{4}{2}\cdot \tfrac{1}{3}(\delta - t)^3 \Big|_0^{\delta}$$

$$= \frac{1}{2}\cdot \tfrac{4}{3}\, \delta^3$$

$$\Rightarrow \quad \tfrac{1}{2} H^2(f_0 \| g) = \tfrac{1}{2}\cdot\tfrac{1}{3}\,\delta^3 \quad \Rightarrow \quad H^2(f_0 \| g) = \underbrace{\tfrac{1}{3}\,\delta^3}_{\epsilon^2}$$

$$\epsilon = \sqrt{\tfrac{1}{3}}\;\delta^{3/2} \qquad\qquad \epsilon^2 = \frac{1}{4n}$$

$$\Rightarrow \quad \underbrace{\delta}_{\omega} = \underbrace{(\sqrt{3})^{2/3}\cdot \epsilon^{2/3}}$$

$$n\, H^2(f_0 \| g) = \frac{n}{3}\cdot \delta^3 = \frac{n}{\cancel{3}}\cdot \underbrace{\epsilon^2}_{\frac{1}{4n}}\cdot \cancel{3} = \tfrac{1}{4}$$

$$m \geq \frac{1}{2}\,\Phi(\delta')\,[\,1 - \underbrace{\|P_f^n - P_g^n\|_{TV}}_{1/2}\,]$$

$$= \frac{1}{4}\,\underset{\text{squared}}{\Phi}(\delta') = \frac{1}{4}\,\delta'^2 = \frac{1}{4}\left(\frac{1}{2}\delta\right)^2 = \frac{1}{16}\cdot 3^{2/3}\cdot\left(\frac{1}{4n}\right)^{2/3}$$

$$= O\left(\underline{n^{-2/3}}\right)$$

Is there any estimator for $f(0)$ that has $O(n^{-2/3})$ error? Yes! $\rightarrow$ an estimator that is Orderwise Optimal

minimax